Counting Techniques

Directions

You are the proud owner of a mathematically themed ice cream parlor that features n flavors. (I suggest playing around with the 'small n' situations to start with.). You sell both cones and bowls of ice cream. Note that cones and bowls provide different eating experiences. For cones, you must eat the scoops in order from top to bottom while bowls do not have this restriction.

For these first sets of questions, we add the additional ice cream parlor rule that when a patron makes an order, they must choose different flavors for each scoop. We will call this the "No Boring Orders Rule". Later on we will remove this rule, but for the questions 1 to 2, this rule is in effect.

Exploration 1

Suppose there are 4 flavors and a patron orders two scoops. How many different cones of ice cream are possible? How many bowls of ice cream are possible? Which has more possibilities cones or bowls? Why? What is the answer if you have 5 flavors and 2 scoops?

Exploration 2

Suppose there are 4 flavors, one of which is vanilla, and a patron orders two scoops. How many different cones of ice cream are possible with one scoop of vanilla? How many different cones of ice cream are possible with no scoops of vanilla? Why must the sum of these two answers be 12?

Exploration 3

Suppose there are n flavors and a patron orders n scoops. How many different cones of ice cream are possible? Try this for small n. Can you construct a stage based process to count the number of cones possible using the multiplication principle. Can you modify this stage based process to count the number of cones with k scoops?

Now let's remove the "No Boring Orders Rule" so that patrons can order multiple scoops of the same flavor.

Exploration 4

Suppose there are 4 flavors and a patron orders two scoops. How many different cones of ice cream are possible? How many bowls of ice cream are possible? Which has more possibilities cones or bowls? Why? What is the answer if you have 5 flavors and 2 scoops?

Exploration 5

Suppose there are n flavors and a patron orders n scoops. How many different cones of ice cream are possible? Try this for small n. Can you construct a stage based process to count the number of cones possible using the multiplication principle. Can you modify this stage based process to count the number of cones with k scoops?